

1. Introduction

Environmental issues are among the contemporary conflicts in society and are becoming an increasingly important topic in sociology. The literature adapts both qualitative and quantitative methods across a wide range, focusing on major actors in environmental conflicts. Researchers examine state discourse and policies to understand how state institutions approach and practice in environmental conflicts. State discourse is one of the main topics of analysis in studying how sovereign powers engage, shift, neutralize, and manipulate the common discourse for an event in society. Many studies focus solely on government officials' public statements during crises.

Although it is important, a longitudinal data source that includes opposing opinions would allow us to track shifts in the discourse. Parliamentary speeches provide a perfect opportunity in such regard. Parliament, with government and opposition party MPs, including some environmental activists or ecologists, serves as a hub for national political discourse data. The aim of this research is not just to understand how the government is changing the discourse, but also to examine how other members of parliament participate or do not.

Environmental conflicts in Turkey are studied distinctly in the literature, regardless of the theoretical bases of neoliberal developmentalism(Adaman et al, 2018; Adaman et al., 2016; Erenşü, 2016), populism(Akbulut&Adaman, 2021; İnal, 2022), labor(Çelik, 2024, 2021), or extractivism(Canefe, 2016; Üner, 2023). Furthermore, state discourse on some environmental conflicts is well documented (Özen, 2022, 2024), and the state's actions and speeches are described in terms of polarization (Özler & Obach, 2018) and criminalization (Kurtiç, 2022). More broadly, the importance of environmental conflicts, both for the state and the public at large, is examined through the question of whether environmentalism is an elite struggle (Bosworth, 2022; Morrison & Dunlap, 1986).

In addition to the literature that uses qualitative and quantitative methods (See: Uluğ & Cohrs, 2017; Taniyıcı, 2010; Aykaç, 2023; Konak Ünal, 2021). To describe environmental conflicts and state discourse in Turkey, there are some research projects that dealt with a large corpus of parliamentary speeches in Turkey by using computational methods: Both studying with the ParlaMint TR-Corpus to diversify the sentiments in the parliament with machine learning models(Kurtoğlu Eskişar& Göltekin, 2022) and studying gender specific political agendas with topic modelling(Toplu et al., 2024). Therefore, studying environmental politics discourse in the Turkish parliament with big data would provide an opportunity to observe and measure discursive shifts using more extensive data sources, temporally and relevant to political party members. Both in the field of computational social sciences and in environmental studies conducted in Turkey, there is a need for research that examines changes in parliamentary discourse using large datasets and reveals general norms and shifts.

This research aims to visualize general patterns in parliamentary environmental debates and to reveal the connections between shifts in parliament and environmental movements in Turkey. Additionally, the study will examine the proportion of environmental discussion in the opening speeches and the parties' speaking time.

2. Research Questions and Hypothesis

1. How are environmental debates constructed in the Turkish Parliament?
 - 1.a. How environment environment-related debates in TGNA changed through time?
 - 1.b. Which parties are more involved in environmental issues in the parliament?

- 1.c. Whether the frequency of these debates coincide with the environmental catastrophes in Turkey?
- 1.d. How much time political parties dedicate to environmental issues compared to their total speaking time?

The study proposes the following three hypotheses, grounded in the literature and in theory, yet applicable to measurement using computational methods.

H1: JDP is more likely to frame environmental issues through development, innovation, infrastructure, and welfare than opposition parties.

H2: Environmental justice-oriented discourse in parliamentary debates increases after grassroots/social mobilizations, but eventually declines.

The next section discusses the choice of the dataset and why it is appropriate for these research questions and hypotheses.

3. Data

This study will use two already-collected datasets: ParlaMint and EJAtlas; it will not perform web scraping from any platform. ParlaMint, an open-source database by Clarin Erin and EJAtlas, is a public geographical atlas created by a collective of political ecology scholars (Temper et al., 2018). The primary dataset, ParlaMint (Erjavec et al., 2025), is a linguistically annotated, multilingual, and comparable corpus of parliamentary debates. The dataset includes data from 29 countries. This study will use ParlaMint TR-Corpora, which is available from the project's GitHub repository. The project defines TR-Corpus in the following way: "The current version of the corpus contains transcripts of debates in the last four terms (from 24 to 27) of the Grand National Assembly of Turkey, which is approximately 50 million words recorded in 1341 sessions from June 2011 to December 2022." (2025).

4. Methodology

In this section, we will examine why this research chooses computational methods to implement it. Computational social science is an interdisciplinary field of the social sciences and STEM that seeks to understand human behavior by applying computational techniques to large datasets from social media sites, the Internet, and other digital archives (Edelmann et al., 2020). Today, digital archives include an unprecedented amount of textual data, including governmental documents, transcripts of parliamentary speeches, and legislative debates (Bail, 2014). Computational social sciences includes many researchers trained in computer science or social sciences, as these two branches of science have different methods and theoretical basis to create a combined and comprehensive effort with the strength of the both side is an important task for the CSS scholars (Salganik, 2019; Edelmann et al., 2020; Bail, 2014; Hofman et al., 2021).

As a methodology, dictionary methods would be used. Dictionary approaches, although among the oldest methods in text analysis, are transparent and simple compared to more novel methods. Also, like supervised and unsupervised learning, dictionary methods can reduce complex data sources by classifying words into categories (Grimmer & Stewart, 2013). The relevance of dictionary methods in conflict research (Baden & Tenenboim-Weinblatt, 2018) and

in the study of hate speech and conflicts on social media (Ben-David & Matamoros-Fernández, 2016; Cohen et al., 2014) is evident.

However, this study does not use LDA or any other Topic Modelling methods because, in such research, the frames/topics would be determined by analyzing a large corpus. Here, the theoretical framing is already decided (extractivism, developmentalism, environmental justice, energy, and sustainability). This study examines how these frames are incorporated into parliamentary speeches, given their conceptualization as the major terms in the environmental sociology literature. Unsupervised methods lack predetermined categorization and a priori-defined theoretical frameworks. They operate inductively, with few modeling assumptions, and discover latent features of texts (Maerz & Pushmann, 2019; Lucas et al., 2015; Grimmer & Stewart, 2013). Topic modelling can be useful for studying a massive data source and classifying the dynamics of texts; however, researchers argue that it can be criticized for being merely descriptive and exploratory (Törnberg & Törnberg, 2016) rather than testing causal relationships (Chen et al., 2023), similar to dictionary methods. For the logistics and these being the initial steps of the project, we will only use dictionary methods, transparent, simple, and well-established.

5. Research Design

Then, ParlaMint data will be separated for each of those dates based on the post- and pre-protest periods in parliament. The ParlaMint data will be checked six months before and after each event and filtered using the defined environmental_dictionary. This dictionary, coded in Python, will include subsets such as Extractivism = {maden, taş ocağı, kalkınma, gelişim, altın, rezerv, ruhsat, madencilik}. These keyword dictionaries can be updated based on the texts mentioned in the protest event dataset. For example, if the protest event is anti-HEPP movements in the Black Sea region, the keyword dictionary can be defined as follows: Anti-HEPP = {baraj, HES, hidroelektrik, santral}. However, data filtering will initially start with more general terms such as energy, development, extractivism, environmental justice, sustainability, and political ecology. Each pre-event, event, and post-event period will be filtered and labeled with neutral, grassroots, and developmentalist sub-discourses, and the prevalence of speeches will be aggregated and counted across the timeline.

Keyword Dictionary:

```
ENV_KEYWORDS = {  
    "extractivism": [  
        "madencilik", "maden", "altın madeni", "taş ocağı",  
        "kömür", "linyit", "rezerv", "ruhsat", "altın"  
    ],  
    "energy_projects": [  
        "hes", "hidroelektrik", "termik santral",  
        "nükleer santral", "enerji yatırımı",  
    ],  
    "developmentalist": [  
        "kalkınma", "büyüme", "yatırım",  
        "istihdam", "milli ekonomi", "milletimizin refahı",  
        "dışa bağımlılık"  
    ],  
    "sustainability": [  
        "çevre", "doğa", "ekosistem",  
        "biyoçeşitlilik", "iklim değişikliği",  
        "sürdürülebilirlik", "politika yapıcılar"  
    ],  
    "environmental_justice": [  
        "çevre adaleti", "yaşam alanı",  
        "yerel halk", "hak mücadelesi", "direniş",  
    ],  
}
```

Figure 1

Figure 1 defines the keyword dictionaries in this analysis, which are divided into 5 parts and include relevant keywords within the theoretical frames. After applying these filtering and keyword-dictionary methods, two categories of consequences will be available. First, without using EJAtlas protest data, we can see which political party MPs discuss environmental issues more, and how MPs' speeches are clustered by topic and party affiliation. This data would also be ratioed by the number of speeches and members each party has, so it will not fall into the pitfall of parties with more MPs not being counted more. The other type would be checking the dates of protests in EJAtlas and examining the use of environmental keywords around those dates.

By this filtering, we can test our hypothesis. For H1, we will examine the mean probability of the frames “developmentalism” and “energy” for each party and compare them. As well as comparing the rates of “extractivist” and “environmental justice” frames in each party. If the government is leaning more towards developmentalism and energy discourse, it must be tested how much the rate of these two frames compares to the rest when looking only at government MP speeches. The mean probabilities of developmentalism and energy frames will also be compared with those of opposition parties.

For H2, we will look at the prevalence of all frames in the pre- and post-protest periods. We will also look at the shifts of “environmental justice” frameworks (especially in the left-wing parties) and the increase in “energy” and “developmentalism” in the government and other right-wing parties. Since Parla-Mint data includes time, party ID, and each party's political orientation, this will be feasible.

For H3, we will examine the periods without major protests and measure the prevalence of frames during those periods. Measuring the “sustainability” frame during that period will also be important, as it is a neutral frame compared to others.

6. Results

[46]:

categories	developmentalism	energy	environmental_justice	extractivism	sustainability
year					
2011	882	632	25	969	125
2012	2379	1806	89	3084	267
2013	2391	1983	189	3202	347
2014	2378	2167	141	3656	303
2015	1320	1032	124	1985	123
2016	3559	3015	283	4985	430
2017	2487	1867	191	3350	333
2018	2622	1768	166	3258	356
2019	2642	1764	128	3484	458
2020	2735	2174	198	4225	440
2021	2823	2167	153	3468	533
2022	2442	1914	155	3288	456

Figure 2

Figure 2 shows the number of speeches that are categorized with the keyword dictionary in Figure 1. As can be seen, extractivism is the most prevalent topic in parliament speeches, and there is a steady focus on developmentalism and energy in these speeches. Interestingly, we see an increase in all three frameworks in 2016, with each, except sustainability, reaching its peak. Also, we see sustainability keywords becoming increasingly common since 2016.

[43]: <Axes: title={'center': 'Share of Energy Discourse in Parliamentary Environmental Debates'}, xlabel='Year', ylabel='Proportion of Speeches'>

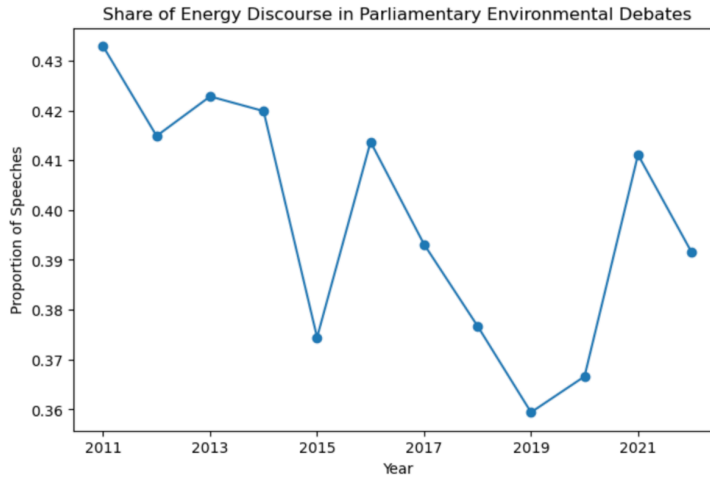


Figure 3

Figure 3 shows an increase in the prevalence of energy keywords from 2015 to 2016 and from 2020 to 2021. There are periods of increase and then decrease in the energy debates.

```
[52]: categories
[extractivism]                23733
[developmentalism]            16154
[energy]                      11553
[extractivism, developmentalism]  5383
[extractivism, energy]         4260
[extractivism, energy, developmentalism] 2535
[energy, developmentalism]      2153
[sustainability]               941
[environmental_justice]        729
[extractivism, energy, developmentalism, sustainability] 661
[extractivism, developmentalism, sustainability] 620
[developmentalism, sustainability] 586
[extractivism, sustainability]  571
[extractivism, environmental_justice] 418
[extractivism, energy, sustainability] 287
[energy, developmentalism, sustainability] 207
[extractivism, energy, environmental_justice] 186
[energy, sustainability]        165
[extractivism, developmentalism, environmental_justice] 99
[energy, environmental_justice]  95
Name: count, dtype: int64
```

Figure 4

Figure 4 shows that from all combinations in the parliamentary speeches, extractivism, energy, and developmentalism keywords are by far the most used. More so, their various combinations follow them. After sustainability, we have the environmental justice dictionary. Although this is an expected outcome, the difference between the numbers is surprisingly high. Also, there is a certain number of speeches in which environmental justice is counted alongside extractivism and developmentalism, or in which both are explicitly addressed and checked.

```
[105]: df[ENV_KEYWORDS.keys()].sum()
```

```
[105]: extractivism      40362
energy_projects      22299
developmentalist     27079
sustainability       33328
environmental_justice  1918
dtype: int64
```

```
[106]: env_cols = list(ENV_KEYWORDS.keys())

df["env_speech"] = (df[env_cols].sum(axis=1) > 0).astype(int)
```

```
[107]: df["env_speech"].value_counts()
```

```
[107]: env_speech
0      652365
1       87379
Name: count, dtype: int64
```

Figure 5

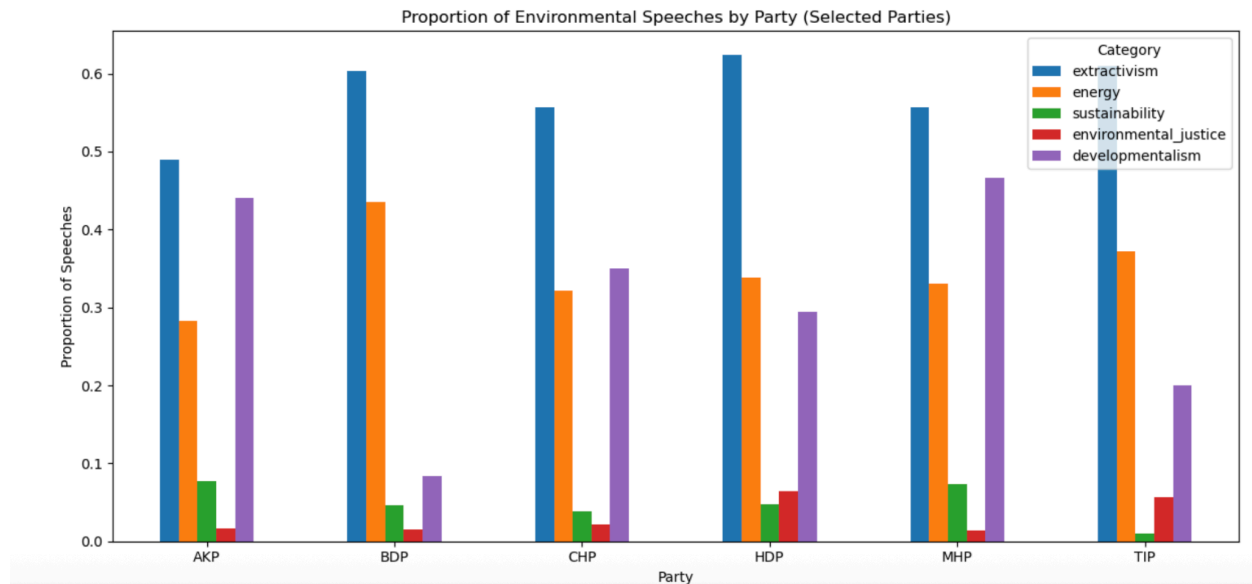
The numbers in each subcategory do not add up to the total environment speech due to overlaps in some speeches. Some speeches can contain both extractivism and developmentalist keywords. That means environmental discourse is spoken through intersecting frames with overlaps. 87,379 environmental speeches out of 739,744 total speeches, which adds up to 11,8% of the total parliamentary speeches.

```
[61]: <Axes: title={'center': 'Gezi Park-Related Parliamentary Speeches (2011-2022)'}, xlabel='Year', ylabel='Number of Speeches'>
```



Figure 6

Figure 5 is an example of measuring Gezi protest-related keywords in the parliament data. This is just an example of how protest data from EJAtlas can be correlated with parliamentary protest-related speech. It is possible to see the temporal prevalence of protest events in the data we have. Dictionary methods are useful because protests that do not include specific keywords can still be counted, minimizing the risk of counting irrelevant speeches by using common vocabulary.

**Figure 7**

Lastly, in Figure 6, we can see which parties are talking with which keywords. Although we can see they are using more extractivism keywords, we cannot know at this point whether they are using positive or negative connotations. We cannot know whether parties use similar vocabulary because of the conversation they are having, although they are contradicting each other. However, there is some insight in this graphic as well. Sustainability keywords seem to be used more by AKP and MHP MPs, which can be explained by their use to legitimize themselves as environmentally sensitive.

If we look at the difference between neutral terms of sustainability and left-leaning terms of environmental justice, we also see some shifts. HDP and TIP use the environmental justice terms more frequently than the sustainability discourse. This is a result parallel with what the literature suggests. However, the difference between HDP and TIP is interesting, given that the Kurdish movement has included political ecologists and environmental activists, and has had these people as MPs.

Although DEM is not available in this data because it was founded in 2023, and YSP has existed for only a brief period, we can still see changes among different political parties within the Kurdish movement in parliament. Comparing BDP and HDP shows that, together with TIP, these three parties deliver more speeches on extractivism, energy, and environmental justice than other parties. HDP's use of environmental justice keywords more than BDP parallels the greater embrace of environmentalism in the Kurdish movement as well.

7. Ethics and Limitations

Two datasets are used in this study, neither of which contains private information about individuals. ParlaMint is an open-source data platform that combines public parliament documents, and EJAtlas provides cases of environmental justice struggles that are displayed in public newspapers or websites. Any private information, names, ID numbers, permanent residence, or sensitive information of the subjects(race, religion, or sexual orientation) is not included in the datasets. Therefore, informed consent is not required for the research.

Even though this is a large-scale analysis, the results are descriptive rather than causal. The hypothesis will show the possibility of relations in the parliamentary discussions. Limitations of the methods could result in missing some speeches or counting irrelevant speeches due to the defined keywords and framing. To justify the research design and move beyond dictionary methods, the methodology can be further strengthened by using LLM-assisted deductive coding with supervised learning.

8. Conclusion

This study aims to understand shifts in environmental discourse in the Grand National Assembly of Turkey (TGNA) by using dictionary methods. The research analyzes two data sources, ParlaMint Tr-Corpus and, as a secondary use, EJAtlas protest event data. The study aims to fill a gap in the literature by analyzing environmental discourse shifts in parliament using computational methods across a large corpus.

9. Bibliography

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